Part 1: UI, Camera & The Gumball 00:00

Camera Controls

- Drag Orbit View
- Shift + Drag Pan View
- Ctrl + Drag (or Scroll) Smooth Zoom
- Zs Zoom Selected (centers view on object)

Viewport Management

- Dbl-Click View Name Toggle max/4-view
- Click View Name Change display (Shaded, etc)
- F7 Toggle visibility of the Grid

Gumball Tool (Widget for transforms)

- **Drag Arrows** Move object along axis
  - + Tap Alt Duplicates object (Copy)
- **Drag Arcs** Rotate object
  - + Hold Shift Constrains to Ortho (90° snaps)
- **Drag Squares** 1D Scale (stretches 1 axis)
  - + Hold Shift Uniform 3D Scale
- **Modify Gumball**
  - Dbl-Click handle Move Gumball only
  - Click Menu icon Toggle "Auto Reset"

Visibility & Selection

- L-R Drag Window select (fully enclosed)
- R-L Drag Crossing select (anything touched)
- Shift + Click Add to selection
- Ctrl + Click Remove from selection
- Invert + Hide / Lock Isolate/Lock items

Precise Moving

- Move Move from Pt A to Pt B via Snaps
- Orient3Pt Align 3 source points ➔ 3 target points

Part 2 & 3: Geometry & Sub-Selection 00:16

The Geometry Hierarchy

- Join Combines touching curves/surfaces
- Explode Breaks Polysurfaces into parts
- ExtrudeCrv Pulls 2D lines into 3D geometry
- Cap Closes flat holes to create a solid
- PlanarSrf Fills flat 2D curve with a surface
- DupBorder Extracts the outline of a surface

Control Points & Sub-Select

- F10 (or PointsOn) Show editable points
- F11 (or PointsOff) Hide editable points
- Ctrl + Shift + Click Select 1 face/edge of a solid
- ConvertExtrusion Fixes simple extrusions for F10

NURBS vs Meshes vs SubD

- NURBS Smooth, exact math geometry (CAD default)
  - Rebuild Changes Point Count & Degree
- Meshes Polygonal geometry (Vertices/Edges)
- SubD Organic, clay-like geometry
  - Tab Toggle Smooth/Box mode
  - Ctrl + **Gumball Drag** Extrudes SubD face
  - Subdivide Breaks 1 SubD face into 4

Part 4, 5 & 6: Layers, Submenus & 2D to 3D Forms 00:36

Layer Management

- Tab Press while renaming to quickly create next layer
- Click Layer ➔ `Change Object Layer`
- **Sub-layers** Drag & Drop. Prefix with "Z_" to sort to bottom.

Curve to 3D Generators

- `Loft` Surface bridging multiple curves
  - Options: Normal, Straight Sections, Align Seams
- `Revolve` Spins curve around axis (Full Circle)
- `Pipe` Turns curve into a tube (Cap: Flat/Round)
- `Sweep1` Extrudes shape along 1 guide rail
- `Sweep2` Extrudes shape bridging 2 guide rails

2D Editing & Toolbar Submenus

- Click **Corner Triangle** Expands nested toolbars
- `Line` ➔ Perpendicular/Tangent to 2 Curves
- `Offset` Expands/shrinks a curve evenly
- `CurveBoolean` Rapid trim: Click inside regions to keep
- `FilletCorners` Rounds all 2D corners at once
- `BlendCrv` Smoothly connects 2 curves (G3/G4)
- `InsertControlPoint` Adds editable points

Part 7 & 8: Snaps, Aids & CPlanes 01:06

Nudge (Micro-movement)

- `Arrows` Normal Nudge (0.2 units)
- `Ctrl` + `Arrows` Small Nudge (0.05 units)
- `Shift` + `Arrows` Large Nudge
- `PageUp` / `PageDown` Vertical Nudge (Z-axis)

Snaps & Constraints

- `Shift` Temporarily toggles Ortho (45°/90°)
- `Alt` Temporarily toggles Object Snaps on/off
- **Planar Snap** Keeps drawing flat to first clicked elevation
- **Osnap Types:** End, Near, Pt, Mid, Cen, Int, Quad, Knot

Construction Planes (CPlanes)

- Top=XY, Front=ZX, Right=ZY. Commands act relative to view.
- `Trim` ➔ Apparent Intersections = Yes (Allows cutting visually overlapping curves from specific views)
- `Project` Throws curve onto 3D surface (View dependent)
- `ProjectToCPlane` Flattens 3D curve absolutely flat

Part 9 & 10: Advanced Solid Edits 01:20

Shortcuts & Customization

- `Space` / `Enter` / `Click` Repeats last cmd
- `Click` Opens Quick Menu
- `Ctrl` + `Drag` Saves toolbar button to MMB Menu
- `SelCrv` , `SelSrf` , `SelLast` , `SelDup`

Splitting, Scaling & Booleans

- `Rotate3D` Rotates around a custom drawn axis line
- `Scale1D` / `Scale2D` Stretches 1 or 2 axes via points
- `InterpCrvOnSrf` Draws curve snapped directly on surface
- `Split` vs `Trim` Split leaves pieces; Trim deletes excess
- **Booleans** (Requires closed solids)
  - `BooleanUnion` Fuses overlaps together
  - `BooleanDifference` Carves holes
  - `BooleanIntersection` Keeps only the overlap
- `FilletEdge` / `ChamferEdge` Rounds/bevels solid edges

Part 11: Arrays, Flow & Cage Deformations 01:49

Repeating Geometry (Arrays)

- `ArrayLinear` / `Array` / `ArrayPolar` Duplicates in rows, 3D grids, or circular patterns
- `ArrayCrvOnSrf` / `ArraySrf` Patterns objects onto specific targeted surface coordinates

Global Flow & History

- `FlowAlongSrf` Warps a flat object to wrap identically over an organic surface
- `History` (Record=Yes, Lock=Yes). Modifying the parent curve automatically updates the generated child geometry.

Deformations

- `Twist` / `Bend` / `Taper` Applies global, sweeping bending or twisting forces along an axis
- `CageEdit` Wraps objects in an editable point-box (BoundingBox > World > 4,4,4) to organically squish/stretch groupings